

METRO TICKET GENERATION SYSTEM

Overall Project Report

ServiceNow • Phases 1–5 • Requirements to Final Presentation

Consolidated project summary with implementation evidence and screenshots

1. Project Overview & End-to-End Architecture

The Metro Ticket Generation System is developed in ServiceNow as a phased solution that moves from requirement analysis and planning through backend configuration, passenger-facing UI/UX, testing and security, and final deployment/documentation.

Five-phase delivery structure

Phase	Primary focus	Outcome
1	Requirement Analysis & Planning	Approved blueprint, scope, MVP, rules and roadmap
2	Backend Development & Configurations	Data layer, tables, fields, relationships and master data
3	UI/UX Development & Customization	Booking catalog, dynamic fare logic and QR experience
4	Data Migration, Testing & Security	Validation, QA, ACL review, data integrity and field mapping
5	Deployment, Documentation & Final Presentation	Troubleshooting, documentation, demo and future roadmap

Architecture

User → Service Catalog → Variables → Client/UI Logic → Validation → Automation / Field Mapping → Backend Table → Ticket / QR Experience

Phase 3 connects the passenger-facing catalog to the backend foundation. Phase 4 adds validation, security and field mapping, while Phase 5 prepares the solution for deployment, handoff and presentation.

2. Phase 1 — Requirement Analysis & Planning

Phase 1 establishes what the system must achieve before development begins. Its focus is requirement discovery, clarification, prioritization and planning. The planned journey is Passenger → Journey Details → Validation → Fare Calculation → Review → Confirmation → Ticket Generation → Notification → Ticket History.

MVP scope

- Station selection and source/destination validation
- Journey date and passenger-count validation
- Fare calculation
- Booking creation and unique ticket generation
- Ticket status and ticket viewing
- Confirmation notification

Out of initial scope

Real-time train tracking and external transit APIs, online payment gateway and financial settlement, QR gate scanning / smart-card integration, dedicated mobile application and advanced SMS/push integrations, and AI-based route optimization or dynamic pricing were identified as future/out-of-scope items.

Planning risks and roadmap

Key risks include unclear fare policy, changing requirements, duplicate tickets, incorrect access, poor master data and integration dependencies. The planning material proposes business approval, scope baselining, unique identifiers, early ACL definition, master-data validation and explicit MVP boundaries.

Phase 2 Evidence — Metro Station Details Table Configuration

dev423779.service-now.com/nw/nav/ui/dasid/params/target/sys_db_objed.do%3Fsys_id%3Df650290c2f1b475005ea298a6fa4e36e%26sysparm_record_target%3Ds...

servicenow

AllFavoritesHistoryWorkspaces

Table - Metro Station's Details

Search

DeleteUpdateDelete All Records

Table Metro Station's Details

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* LabelMetro Station's Details

ApplicationGlobal

* Nameu_metro_station_s_details

Remote Table

ColumnsControlsApplication Access

Table Columnsfor textSearch

1 to 7 of 7New

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)		32	false
Updates	Integer	(empty)		40	false
Updated	Date/Time	(empty)		40	false
Updated by	String	(empty)		40	false
Created by	String	(empty)		40	false

Phase 2 Evidence — Station Record Creation and Verification

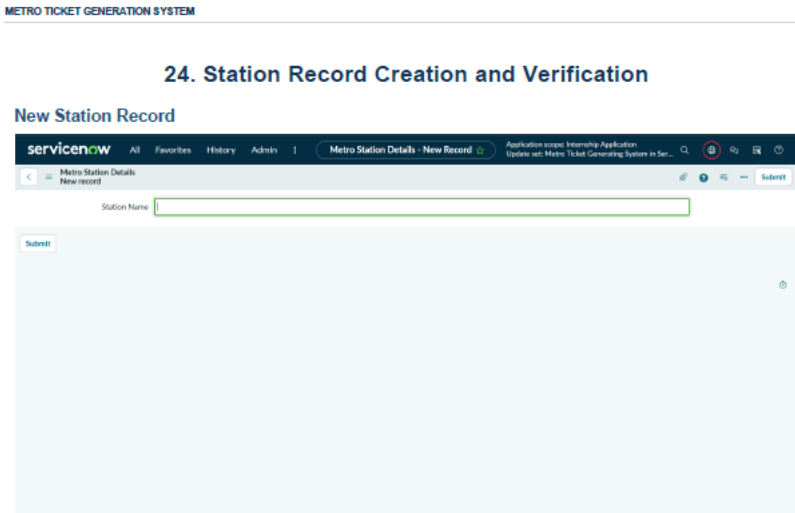


Figure 24. Metro Station Details — New Record form showing the Station Name field used for station master-data creation.

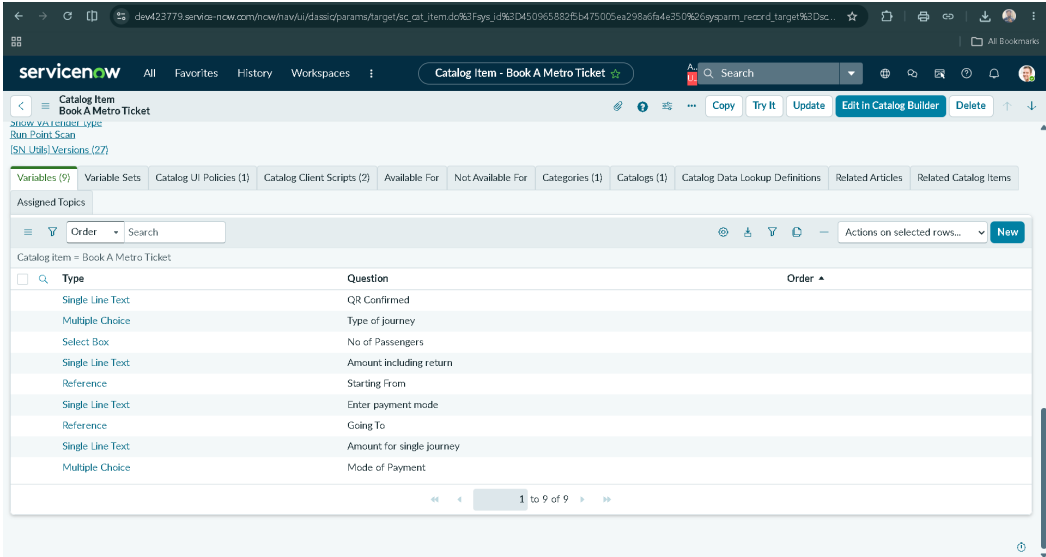
Station Records List



Figure 25. Metro Station Details list showing the configured station records, including Uppal Stadium, Panjugutta, Madhapur, L B Nagar, Kukatpally, Jubilee Hills and Ameerpet.

Together, these screens show the practical backend flow: the Station Name field is presented for record creation, and the saved station records can then be viewed in the Metro Station Details list. This supports the Phase 2 testing objective that valid station records can be created and retrieved successfully.

Phase 3 Evidence — Catalog Item Variables



Phase 3 Evidence — Book A Metro Ticket Configuration

servicenow All Favorites History Workspaces Catalog Item - Book A Metro Ticket

Build and modify items faster with the improved [Catalog Builder](#).

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name

Application

Catalogs

Category

State

Checked out

Owner

Secondary owners

Active ☒

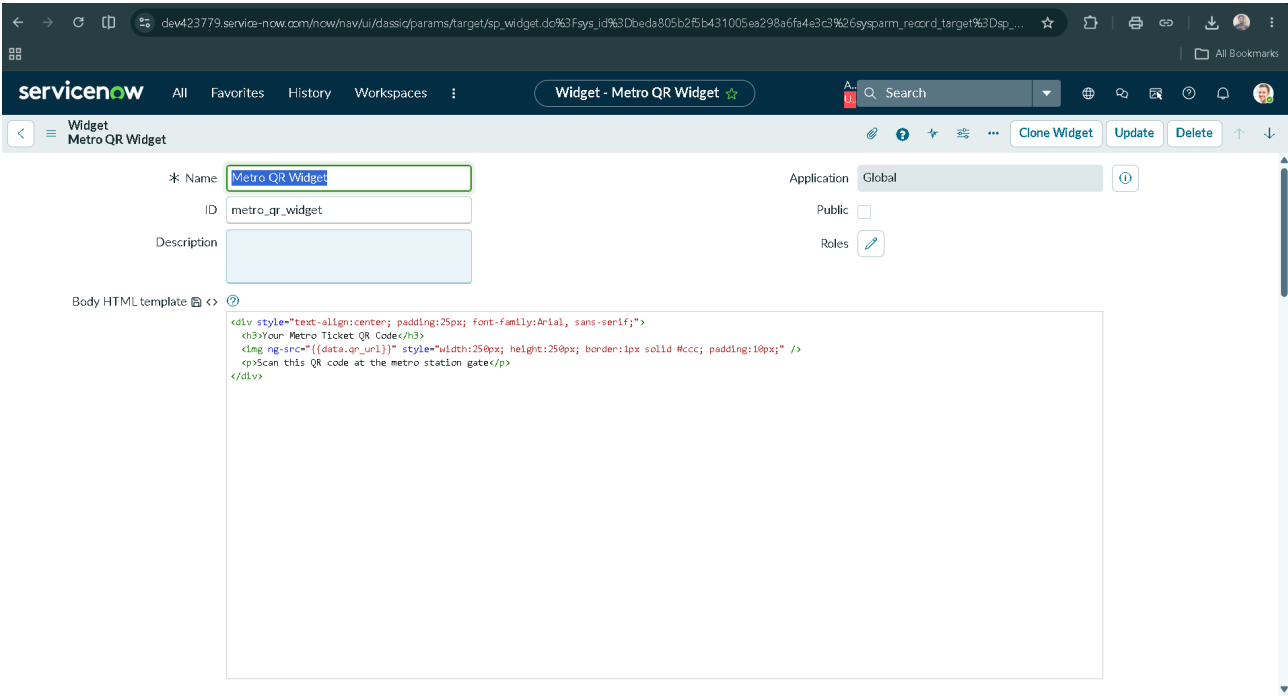
Fulfillment automation level

Item Details Process Engine Picture Pricing Portal Settings

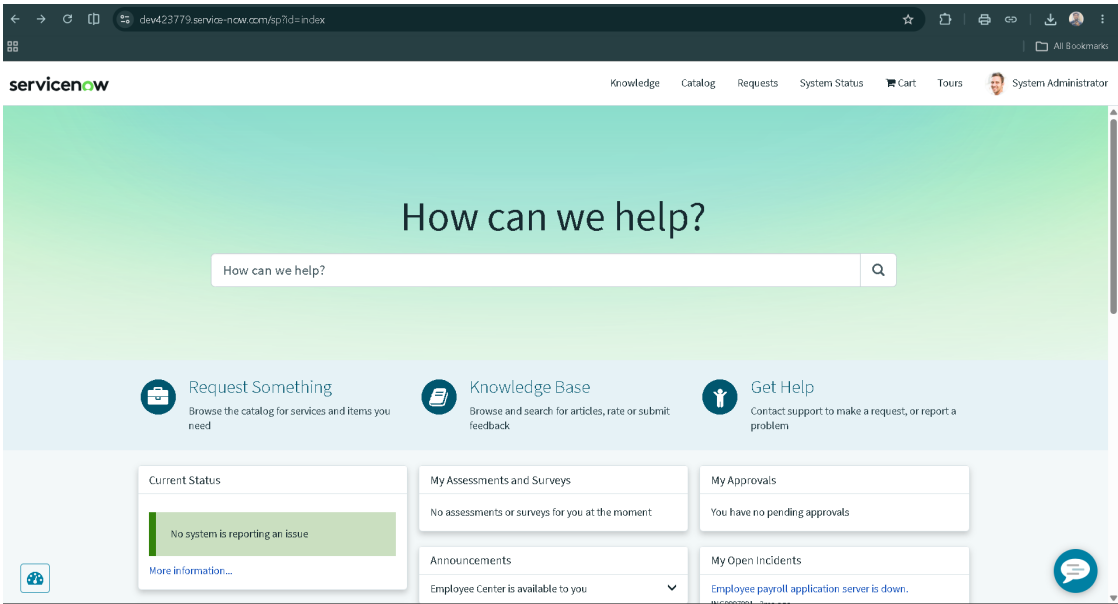
Short description

Description

Phase 3 Evidence — Service Portal Entry Point



Phase 3 Evidence — Passenger Booking Form



Phase 3 Evidence — Fare and Payment Section

dev423779.service-now.com/sp?id=sc_cat_item&sys_id=450965882f5b475005ea29836fa4e350

KnowledgeCatalogRequestsSystem StatusCartToursSystem Administrator

Home > Service Catalog > Office > Services > Book A Metro Ticket

Search Catalog

Book A Metro Ticket

* Indicates required

* Type of journey

☐ single journey

☐ Return journey

☒ single journey

No of Passengers

1

Amount including return

* Starting From

* Going To

Amount for single journey

Quantity: 1

Delivery Time: 2 Days

Add to Cart

Save as Draft

Order Now

Required information

Starting From

Going To

Phase 3 Evidence — Metro QR Widget

dev423779.servicenow.com/sp?id=sc_cat_item&sys_id=450965882f5b475005ea298a6fa4e350

☆ | 📄 | 🖨️ | 🔍 | 📌 | 📱 | 📧 | 📧

All Bookmarks

servicenow

KnowledgeCatalogRequestsSystem Status🛒 CartTours👤 System Administrator

Home > Service Catalog > Office > Services > Book A Metro Ticket

Search Catalog 🔍

* Starting From

* Going To

Amount for single journey

Mode of Payment

☒ Upi

☐ Card

☐ Others

Add attachments

📁

Choose a file or drag it here.
Copy and paste clipboard files here.

Quantity: 1

Delivery Time: 2 Days

🛒 Add to Cart

📄 Save as Draft

Order Now

Required information

Starting From

Going To

🏠

💬

Phase 4 Evidence — Calculated Fare and QR Ticket

22. Test Evidence — Calculated Fare

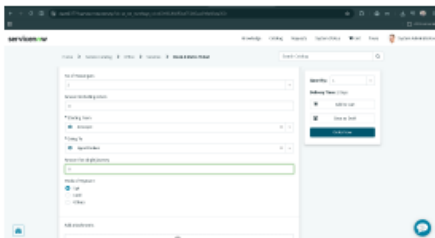


Figure 2. Catalog form showing dynamically populated fare fields.

The screenshot demonstrates the fare fields populated in the catalog interface. During QA, the value should be checked against the configured route, journey type and number of passengers.

23. Test Evidence — QR Ticket



Figure 3. QR ticket modal displayed during the checkout flow.

The QR modal provides the final user-facing ticket experience. QA should confirm that it opens at the intended submission point, displays a valid QR image and does not bypass required validation or authorization.

24. Phase 4 Exit Criteria

- Access controls are reviewed and tested for intended roles.
- Catalog users can complete approved business actions without unnecessary database access.
- Mandatory and validation rules prevent incomplete or invalid submissions.
- Fare calculation is consistent across supported routes and journey types.
- UI policies and client scripts behave correctly after value changes.
- Field mapping and Process Automation populate the intended backend values.
- QR-ticket generation works through the intended checkout flow.
- Positive, negative and boundary QA scenarios are documented and passed.
- Data Integrity checks confirm consistent master and transactional data.
- The solution is ready for Phase 5: Deployment, Documentation & Final Presentation.

25. Final Outcome

Phase 4 establishes the Metro Ticket Generation System as a validated, controlled and secure ServiceNow solution. The combination of ACL review, QA testing, validation, auto-population and field mapping ensures that the user experience is reliable while backend data

Phase 5 Evidence — Supplied Project Screenshots

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17. Visual Evidence — Supplied Project Screenshots

The supplied screenshots are included as implementation evidence for the final documentation and presentation.

Phase 5: Deployment, Documentation & Final Presentation

Troubleshooting

Adherence to Timelines

Innovation

Document: Functional Overview

Document: Technical Blueprint

Document: Setup Manual

Project Demo Video Planning

Explanation Clarity

Resilience & Quality

Scalability & Future Plan

Figure 1. Supplied implementation evidence.

Troubleshooting

Help

Feedback

Assigned to: Kathleen Rose Kelly Tadi

Description

- Debugged DB encoding issues and flow visualisation using logs and print statements to exclude. Reforms and verify variables within database systems and configurations.
- Verified Population Feature: Document misaligned or missing data third-party setup.
- Customized print logs for dynamic operation.
- Performance Testing: Confirmed 10 persons and priority login via Service Portal home beta and previous builds.

Resources

No resources available

Figure 2. Supplied implementation evidence.

Assigned to: Kathleen Rose Kelly Tadi

Description

For each criteria for each milestone (Form, Logic, GE, UAT) and Executed project based on the 4-phase roadmap:

1. Criteria Creation
2. Form Setup
3. Test up & Security
4. Deployment

Resources

No resources available

Figure 3. Supplied implementation evidence.

18. Explanation Clarity

Documentation and presentation follow a logical What → Why → How → Outcome structure. Each major step should have a clear heading, labelled evidence and a short explanation of why the configuration matters.

- Explain business purpose before technical configuration.
- Explain dependencies before demonstrating dependent features.
- Use consistent names for variables, tables and automation components.
- Show actual outcomes after configuration, not only setup screens.

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Phase 5 Evidence — Additional Visual Evidence

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19. Additional Visual Evidence

The next set of supplied screenshots provides supporting evidence for configuration, workflow and project presentation.

Assigned to Kostas Rami Reddy Tati

Description

- Implemented end-to-end automation using:
- Flow Designer (no custom script included or business rules)
- Dynamic Task execution
- Client scripts and UI policies used to minimal, essential use
- Ensured maintainability and ease of handoff to ad-hoc or future developers.

Figure 4. Supplied project evidence.

Assigned to Kostas Rami Reddy Tati

Description

- Clear definition of form variables
- User experience flows
- QR mapping process.

Resources

No resources available

Figure 5. Supplied project evidence.

Assigned to Kostas Rami Reddy Tati

Description

- Included QR generation code snippets.
- Flow Designer snippets for key logic paths.
- Flow Designer workflows (screenshots with explanations)
- Variable-to-field mapping logic from catalog form to custom table
- Custom table schema (Metro Database)
- Portal Catalog and Category adding process

Resources

Figure 6. Supplied project evidence.

Assigned to Kostas Rami Reddy Tati

Description

- Step-by-step setup: Catalog → Flow → Script Includes → Modifications → ACLs.
- Provided step-by-step manual for recreating the solution in a new Personal Developer Instance (PDI).
- Catalog configuration
- Variable configuration
- ACL and role configuration
- Test automation data
- Screenshots and order of operations included for clarity.

20. Realism & Quality

The solution reflects a realistic city-metro workflow with station references, passenger-based fare calculation, Journey types, payment selection, backend processing and QR-ticket presentation. Realistic sample routes and data make the demonstration meaningful.

- Use realistic station names and route combinations.
- Validate amounts against the configured fare matrix.
- Test normal and exception paths.
- Ensure the demo reflects the documented implementation.

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Phase 5 Evidence — Presentation and Quality Evidence

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21. Final Evidence & Presentation Checklist

The remaining supplied screenshots are retained as final supporting evidence. They can be referenced during the final presentation to prove implementation and user-flow behavior.

Assigned to: Kostas/Rami Roddy Test

Description

- Walkthrough from partial ticket submission to QR scan by station authority.
- Scope Included:
 - Partial walkthrough: Home → Book A Metro Service → Submit Request
 - User request submission
 - Request status tracking
 - Status showing in Table
- Includes narration for each step and role-based perspective (requester, fulfiller).

Figure 8. Supplied project evidence.

Assigned to: Kostas/Rami Roddy Test

Description

- All tasks above completed with screenshots, clear screenshot naming.
- Documentation and presentation below legend, organized structure
 - What - Why - How - Customer
- Each step includes:
 - Clear headings
 - Labeled screenshots
 - Explanatory notes for reproducibility

Figure 9. Supplied project evidence.

Realism & Quality

View

Download PDF

Assigned to: Kostas/Rami Roddy Test

Description

- Reflects actual city metro workflows including dynamic fare bands.
- Solution tested with realistic use cases and sample data.

Figure 10. Supplied project evidence.

Scalability & Future Plan

View

Download PDF

Assigned to: Kostas/Rami Roddy Test

Description

Scope for SMS ticketing, Email Ticketing, payment gateway integration, and travel analytics dashboard.

Figure 11. Supplied project evidence.

22. Presentation Quality Checklist

- Start with the business objective and user problem.
- Explain the architecture before deep technical details.
- Demonstrate one complete successful booking.
- Show dynamic fare calculation and conditional payment behavior.
- Show the QR-ticket experience.
- Explain troubleshooting and security decisions.
- Finish with measurable outcome and future roadmap.

23. Scalability & Future Plan

- Extend SMS ticketing for confirmations and travel alerts.
- Add Email Ticketing for digital receipts and ticket delivery.
- Integrate a payment gateway.
- Introduce a travel analytics dashboard for route usage, passenger volume and revenue.
- Move route/fare logic toward maintainable data-driven fare tables and server-side calculation.

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